

The IP₃ receptor is ancestrally eukaryotic and has been lost repeatedly outside the animals

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Abstract

The inositol 1,4,5-trisphosphate receptor (IP₃R) is the endoplasmic reticulum's ligand-gated calcium-release channel. Its distribution beyond the animals has been inferred from small samples, and every search for it is exposed to the ryanodine receptors (RyRs), which carry all of its diagnostic Pfam domains. We asked in which eukaryotic lineages an IP₃ receptor gene exists, and answered with instruments that call each record positively as one family or the other and that were benchmarked on 31 decoys before use. Across 763 vertebrate and 6,928 other reference proteomes, the census holds 18,065 records, 8,990 of them IP₃ receptors across 1,402 taxa. The family is present in 662 of the 6,928 non-vertebrate proteomes, across animals, several protist lineages, green algae and early-diverging fungi, and in none of 634 archaeal or 3,537 bacterial proteomes. It is absent from all 384 land-plant and all 1,353 Dikarya proteomes while present in 15 of 48 chlorophyte and 18 of 34 mucoromycote ones. We took 35 clade-level absences to 194 genome assemblies, in which a positive control chosen per clade by measurement demonstrated that the search could recover a long, conserved, multi-exon gene. All 35 absences hold at assembly level. All 99 plant and fungal records were chased individually: 47 are real genes, 52 are fragments, and none is contamination. The family is therefore ancestral to eukaryotes and has been lost independently in land plants, in Dikarya and in several other lineages. Copy number outside the vertebrates runs from zero to eighteen, and the shape of the few structures available separates the two families as the sequence census does.

Introduction

The IP₃ receptor releases calcium from the endoplasmic reticulum when it binds inositol 1,4,5-trisphosphate, the second messenger made when phospholipase C cleaves a membrane phosphoinositide [1, 2]. It is a tetrameric channel of about 2,700 residues per subunit, and in vertebrates it exists as three paralogues, ITPR1, ITPR2 and ITPR3 [3, 4]. Its closest relatives are the ryanodine receptors, calcium-release channels of about 5,000 residues that share its pore and much of its cytoplasmic architecture [5, 6].

Where the receptor occurs outside the animals has been argued from small samples. Comparative surveys of the calcium-signalling toolkit found receptor-like genes in some protists, green algae and early-diverging fungi, and none in the land plants or the yeasts [7, 8, 9, 10]. The land plants are the case that matters most, because plant cells release calcium in response to IP₃, no plant receptor gene has been identified, and the standing review of

the question asks in its title whether the receptor is real [11]. If the family is absent there, whatever performs that physiology is built from other channels [12, 13].

Two problems make that question harder than it looks. The first is the ryanodine receptor. Every Pfam signature diagnostic of the IP₃ receptor, including the IP₃-binding core (PF08709), the RyR and IP₃R homology domain (PF01365), the RIH-associated domain (PF08454) and the MIR domain (PF02815), also occurs in ryanodine receptors. A search for one family therefore returns the other, and a range built from domain annotation would be a range of the two families mixed. Length separates them in well-annotated proteomes, but length is exactly what a fragmentary or unannotated record does not carry.

The second problem is that a database returns records and not genes. “The family occurs in N proteomes” is a statement about what gene callers found, and “the family is absent from land plants” is a claim about DNA that a proteome cannot make on its own. An absence becomes a measurement only when the same search, in the same genome, demonstrably recovers a comparable gene. A positive control is what licenses a negative claim, and range surveys rarely carry one.

We therefore asked one question, in which eukaryotic lineages an IP₃ receptor gene exists, and answered it in four layers, each checked against the one before. First, every record is assigned to the IP₃ receptors or the ryanodine receptors by a positive test, meaning evidence that it is one rather than evidence that it is not the other, and the tests were benchmarked on decoys before any search ran. Second, the family’s profiles were swept over 7,691 reference proteomes spanning Eukaryota, archaea and a genus-stratified bacterial sample, with each negative claim made at two stated sensitivities. Third, every plant and fungal record was chased individually for contamination. Fourth, the absences were taken to 194 genome assemblies, each carrying a positive control selected by measurement for its own clade. We report the family’s range with that scope attached to every claim, and the fold of the structures that exist as an independent check on the family call.

Results

Three independent instruments separate the IP₃ receptors from the ryanodine receptors

Before any search ran, the discovery scorer was benchmarked on 25 positive controls, the three vertebrate paralogues across a species panel plus the invertebrate and non-metazoan single-receptor grade, and 31 decoys chosen to be hard: six ryanodine receptors, which carry all four family signatures, the MIR-sharing O-mannosyltransferases POMT1 and POMT2, fifteen in-band channels and non-channels, and out-of-band giants. The first run called all six ryanodine receptors IP₃ receptors. The fix, used unchanged for the rest of the work, is a labelled-bait identity margin: a candidate is assigned to whichever family’s labelled baits it is closer to, and only when it wins by more than 0.10. The margin never reads a gene name, and length never decides a call. With it, recall is 24 of 25 and specificity 31 of 31, and all 6 ryanodine receptors are rejected. Applied to every record in the census, the margin leaves an empty gap between the two families (Extended Data Fig. 1). The one missed positive, *Drosophila* Itpr, fell 0.008 short of the identity its novelty component needs, which measures the instrument’s worst case: a lone family member with no close sibling in the set.

The search space was then enumerated to exhaustion rather than sampled. Three family signatures (PF08709, PF01365 and PF08454) were walked through InterPro [14, 15] to cursor exhaustion, because the advertised counts are wrong in both directions: PF08709 advertises 12,339 records and serves 12,507, while PF08454 serves 11,890, fewer than it advertises. The union is 15,421 proteins across 1,488 taxa, and only 10,256 carry all three seeds. A census built on PF08709 alone, the signature that names the family, would have missed 2,914 proteins, 758 of them IP₃ receptors across 385 taxa (Extended Data Fig. 2a, b). A search of the public databases by name, the obvious alternative, had returned almost none of the records outside the vertebrates (Extended Data Fig. 3a).

Every enumerated record was called by domain architecture. A record is a ryanodine receptor if it carries any of

four RyR-specific signatures and an IP₃ receptor if it carries the complete five-signature IP₃R architecture and none of them. That rule calls 6,433 IP₃ receptors, 6,807 ryanodine receptors and leaves 2,181 records uncalled. Because the rule never sees a gene symbol, symbols are an independent label to audit it against: it agrees with the symbol on 2,479 of 2,479 decided IP₃ receptor records and 2,492 of 2,492 decided ryanodine receptor records. Length separates the two called families cleanly, but the uncalled records are short fragments where length says nothing, which is why length never makes the call (Extended Data Fig. 3b).

A rule that reads annotation cannot call a partial record, so a second instrument reads residues. Two profile hidden Markov models [16], one built from 34 IP₃ receptor seeds (2,684 match states) and one from 22 ryanodine receptor seeds, assign each target to the profile that beats the other by more than 10 % of its own score, provided the winner clears 30 bits over at least 200 match states. Calibrated against the architecture call before use, with the seeds removed, the profiles agree on 11,875 records and disagree on 1 (Fig. 1). They also call 2,074 records the architecture rule left unassigned as IP₃ receptors and 240 as ryanodine receptors. Merged by a stated rule, census v3 holds 16,039 records: 8,000 IP₃ receptors, 7,432 ryanodine receptors and 605 unassigned (Extended Data Fig. 4).

The profile search misses no receptor inside the proteomes it searched

Swept over 763 vertebrate reference proteomes, 14,414,821 proteins, the profiles declined 12,609 hits as module-only matches, proteins that share one small domain with a channel model, and found 618 targets the InterPro walk had never returned, 188 of them IP₃ receptors. The completeness check was run in the expensive direction. Of the enumerated IP₃ receptor records in swept taxa that the profiles did not report, all 2,787 were looked up in the search database itself: none was in the database and missed. Every one is a UniProtKB entry that its taxon's reference proteome does not contain. Fifteen vertebrate proteomes carry no family record at all, and copy number per proteome rises with the size of the gene set (Extended Data Fig. 4b), which is why every absence claim below is taken to an assembly.

Iterative search gives a third line of evidence. Three jackhmmer runs [17] seeded from a human, a fly and an amoebozoan receptor intersect on 4,785 family records and differ by at most 162; what the seed changes is only the off-family debris (Extended Data Fig. 5). Iteration returned no record the profile pair calls family that one pass had not already returned. The coded kill criterion written for this family's hazard, a rise in the ryanodine receptor share of a round, fired on none of the three runs that drifted, because accreting unrelated proteins dilutes that share rather than raising it. Measured on the finished models, the seven runs separate cleanly, with the drifted ones at an off-family share of 0.81 or more and the rest at 0.34 or less. The round ceiling caught all three drifted runs, so no drifted round entered any count, and moving the same rule onto the off-family share classifies all seven runs correctly (3 of 3 drifted, 4 of 4 clean).

The family is present in 662 of 6,928 non-vertebrate proteomes and in no prokaryote

The same instrument, unchanged, was swept over 6,928 further reference proteomes: 63,144,898 proteins in four eukaryotic groups that partition Eukaryota with the vertebrate set, all 634 archaeal proteomes, and 3,537 bacterial proteomes sampled as the largest proteome per genus. The family separation transfers intact: of 28,137 targets scored by at least one profile, 2,769 were scored by both above the floor, and 1 of those falls inside the no-call band (Extended Data Fig. 6).

Presence spans the animals and several protist and algal lineages. 662 of the 6,928 proteomes carry an IP₃ receptor call, and 45 of the 135 clades swept do (Fig. 2). It is in 292 of 311 arthropod, 106 of 110 nematode and all molluscan, cnidarian and sponge proteomes, and outside the animals in Amoebozoa, Discoba, ciliates, oomycetes, dinoflagellates, haptophytes and green algae. It is in no archaeal or bacterial proteome. The choanoflagellate *Salpingoeca rosetta* carries a full-length ryanodine receptor that is not a profile seed, so

the split between the two families predates Metazoa, as comparative surveys suggested [7, 10]. With the gene models from the genome sweep added, the census holds 18,065 records, of which 8,990 are IP₃ receptors across 1,402 taxa.

Land plants and Dikarya have lost the receptor that their early-diverging relatives retain

The two absences sit beside presences in the same kingdom. None of 384 Streptophyta (land plant) reference proteomes carries a call, against 15 of 48 Chlorophyta. None of 1,034 Ascomycota or 319 Basidiomycota proteomes carries one, against 18 of 34 Mucoromycota and 6 of 16 Chytridiomycota. The family is also absent from all 60 apicomplexan, 29 microsporidian, 27 glomeromycote, 18 mortierellomycote and 35 kickxellomycote proteomes, and from 16 diatom and 12 red algal ones.

Each negative claim has a positive control inside the same search. Every negative was made twice, at $E \leq 1e-5$ with the two full-length profiles and at $E \leq 10$ with those profiles plus the family's four Pfam domain models, because a short domain model can reach a sequence a 2,684-state channel model cannot. At the relaxed sensitivity, PF08709 returns no substantial match in land plants against 26 in Chlorophyta, and none in Dikarya against 16 in Mucoromycota. PF02815, the MIR domain that every eukaryote carries on unrelated proteins, returns 633 substantial matches in land plants and 4,376 in Dikarya. The instrument finds thousands of the promiscuous domain in the proteomes where it finds none of the family-defining one. Iterating a model seeded inside each group added 9 targets the single pass never reported, and the two that sit in an empty lineage are O-mannosyltransferases, the MIR decoy, rather than receptors.

Every plant and fungal record is a real gene or a fragment of one

A handful of records in kingdoms whose model organisms have none is the shape of a contamination artefact, so all 99 plant and fungal IP₃ receptor records, 64 plant and 35 fungal, were chased individually through four lines of evidence: sequence, the record's own provenance, the nearest relative outside its kingdom and the nearest within it. 47 are real genes and 52 are fragments of real genes; none is a contamination suspect and none lacks genome backing (Fig. 3). The contamination test had full power. A genuine deep homologue is 20–40 % identical to its metazoan relatives and an assembly contaminant is 95–100 % identical to one particular animal. The 22 surviving plant records sit at 19.9–39.9 % identity to anything outside their kingdom and the 25 fungal ones at 20.5–33.5 %, and no record of the 99 exceeds 45.8 %.

Thirty-five clade-level absences hold in controlled genome assemblies

A proteome absence is a fact about a gene caller, so we re-asked each one of a genome. We selected 194 non-vertebrate assemblies, 100.4 Gbp, from the 24,596 NCBI eukaryotic reference assemblies [18] less the 6,216 vertebrate ones, by five rules that sample most finely where the negative claims are: one assembly per phylum, one per class in the most-sampled phyla, one per clade that the proteome sweep called empty, anchor organisms with a known answer, and the highest-copy species per class. The third rule re-derives the target list from the presence table and found three absences no summary had named: diatoms (0 of 16), red algae (0 of 12) and Cestoda (0 of 11), a metazoan clade.

The positive control was chosen per clade by measurement. The vertebrate convention of using the ryanodine receptors as proof of search does not transfer, because architecture-level ryanodine receptors occur in 2 of the 6,928 non-vertebrate proteomes and a silent RyR in a land plant is the correct answer. A single fixed replacement also failed: both apicomplexan classes carry a single MIR protein across their 60 proteomes. Six candidate families, each large, deeply conserved and multi-exon, were therefore run over every clade's own proteomes, and each clade takes the one that is actually present. Apicomplexa take the myosin head domain, present in 36 of 36 Aconoidasida and 23 of 23 Conoidasida proteomes; red algae take the SMC N-terminal domain, present

in 12 of 12. The panel carries 108 baits, of which 37 are IP₃ receptors, 16 are ryanodine receptors and 55 are clade controls, and every one of the 56 control clades is filled.

No genome in the sweep is uncontrolled. In 115 genomes the control is recovered and baits from another kingdom-level group align across the same locus, showing the search crosses the divergence a receptor would have to be found across; in 77 the receptor itself was found; in two ciliate genomes the control is partial or absent, and no absence claim rests on either. Against that, **all 35 clade-level absences hold at assembly level** and none fails (Fig. 4). Ascomycota has no receptor in 31 controlled assemblies, Streptophyta none in 25 and Basidiomycota none in 17. Two land-plant assemblies carry only scattered tblastn traces, short of a gene. The genomic thresholds were measured for this scope rather than transferred: sixteen annotated gene spans run from 3,739 to 324,840 bp with a median of 19,435, so the contiguity bar is set per group, and a cluster of alignments is not counted as a gene until it passes rules for split and chained models (Extended Data Fig. 7).

Copy number outside the vertebrates runs from zero to eighteen

Outside the vertebrates there are no paralogue cells to fill, so the question is how many receptor genes a genome carries. Of the 194 genomes, 117 carry none, 43 carry one, and the rest carry two or more (Fig. 5). The flatworm *Macrostomum lignano* carries 18 complete gene models, the ciliate *Stentor coeruleus* 13 and the sponge *Dysidea avara* 8. The vertebrate count of three is not the family's maximum. The genome also corrects the protein side: the green alga *Cymbomonas tetramitiformis*, which contributes twelve of the 22 surviving plant records, carries three complete gene models.

The fold separates the two families where a structure exists

A structural check is independent of every instrument above, because it reads no gene symbol, domain annotation or alignment score. It is also limited by what exists. AlphaFold DB [19, 20] holds a usable model for 20.9 % of the census's protein records, for 9 of the 134 representatives the downstream analyses use, and for 13 of the 5,861 full-length records (0.2 %). It answers an accession with whatever record it holds, which for human ITPR2 is a 181-residue isoform (Extended Data Fig. 8). The structural test therefore rests mainly on cryo-electron microscopy depositions, chosen by seven rules with the family call taken from this census rather than from entry titles.

The comparison scale was calibrated on this panel before anything was called (Fig. 6). None of 81 pairs involving a negative control reaches TM-align's same-fold bar of 0.5 [21]. Two structures of the same paralogue in different conformations score a median of 0.78 against 0.43 for two different IP₃ receptors, so conformation alone moves the score by more than the difference any fold claim here relies on. Scored against both reference families with a 10 % relative margin, all 20 structures the test could call are called as the census calls them, and none of the 3 negative controls is called at all. The test does not reach the deep lineages: 2 of 7 non-vertebrate models clear the same-fold bar, and the rest prefer the IP₃ receptors by about two to one while falling short on similarity, not on length. Model confidence is highest in the IP₃-binding core (median pLDDT 83.9) and lowest in the pore among named domains (71.0), against 69.5 outside annotated domains (Extended Data Fig. 9b).

Discussion

The IP₃ receptor is an ancestral eukaryotic gene. It is present across the animals, in amoebozoans, excavates, ciliates, oomycetes, dinoflagellates, haptophytes and green algae, and in early-diverging fungi, and it is absent from every prokaryote swept. That distribution cannot be explained by a single origin inside the animals, and the ryanodine receptor in a choanoflagellate places the split between the two families before Metazoa [7, 10]. The absences are then losses, and they are independent. Land plants lost the receptor after diverging from the green algae that still carry it [22], and the fungi lost it along more than one branch: Dikarya, Glomeromycota,

Mortierellomycota, Kickxellomycota and Microsporidia are all empty while Mucoromycota and Chytridiomycota are not, a pattern read against the accepted fungal backbone [23] as several losses rather than one. Loss of this kind is an ordinary evolutionary mechanism [24], and several of the empty lineages are ones with reduced genomes or reorganised calcium signalling: the apicomplexans and microsporidia [25, 26], the diatoms and red algae [27], and the tapeworms, which have shed gene families across their biology [28].

For the land plants the result puts a genomic answer under a question the physiological literature has kept open [11]. Plant cells respond to IP_3 , and no IP_3 receptor gene exists in any of 384 land-plant proteomes or 25 controlled land-plant assemblies. Whatever mediates that response is built from other channel families [12, 13]. We cannot say what it is; we can say that it is not a divergent member of this family missed by a domain-based search, because the same search finds the family in the green algae, finds the MIR domain hundreds of times in land plants, and was made at a sensitivity loose enough to recover a single IP_3 -binding domain.

Three design decisions carry the result, and each is visible in the data rather than asserted. The family call is positive at every stage, so a record is an IP_3 receptor because it is closer to IP_3 receptors than to ryanodine receptors and not because it failed to look like the sister family; on the first benchmark run the obvious instrument called every ryanodine receptor an IP_3 receptor. The census counts proteomes and genomes rather than records, so a well-sequenced lineage cannot outvote a sparse one. And every absence at genome level carries a control chosen for its own clade by measurement, which is what turned a fixed control that fired once per apicomplexan class into one that fires in every apicomplexan proteome.

The claims have limits, and they are stated with the numbers. A proteome absence covers the gene sets of 6,928 reference proteomes and a genome absence covers 194 assemblies; a lineage not sampled is not covered. The profiles are labelled by seeds, and the deep branches rest on seven non-metazoan seeds, so a lineage more diverged than any of them is outside what the profile call has been shown to reach. The labelled-bait margin cannot reach one characterised receptor, *Dictyostelium* iplA, which is in the profile seed set for that reason. No green-plant gene span could be measured, so land-plant assemblies are judged against the global median span. The one other full-length ryanodine receptor outside the animals, in *Capsaspora owczarzaki*, is itself a profile seed and is not independent evidence. The fold test confirms the family call where cryo-EM structures exist and does not reach the deep lineages, whose only models are partial and of low confidence. Finally, the vertebrate absences are not addressed here: the vertebrate proteomes that carry no receptor record were leads for a genome sweep, not losses, and whether any vertebrate has lost a paralogue is a separate question with its own controls.

Methods

Scope

The question is asked of three declared spaces. The proteome space is 763 vertebrate UniProt reference proteomes (14,414,821 canonical proteins) and 6,928 further reference proteomes (63,144,898 proteins) in six groups: non-vertebrate Metazoa, Fungi, Viridiplantae, other protists, all archaea, and bacteria sampled as the proteome with the most proteins in each genus [29]. The four eukaryotic groups partition Eukaryota with the vertebrate set, so the denominators add. Canonical proteomes are used because isoform sets add isoforms of genes already counted. Proteomes listed by UniProt but absent from the release tree were excluded and recorded rather than retried. The enumeration space is every UniProtKB protein carrying PF08709, PF01365 or PF08454, walked through the InterPro API to cursor exhaustion. The genome space is 194 non-vertebrate assemblies (100.4 Gbp) drawn from 24,596 NCBI eukaryotic reference assemblies less 6,216 vertebrate ones by the five rules given in Results, with every rule's selections committed.

The family call

The benchmark was run before any search. The discovery scorer was run against 25 positive and 31 negative controls fetched from UniProt (Swiss-Prot preferred), with MAFFT [30] alignments whose return code was checked, since a silent MAFFT failure degrades to a star alignment. The labelled-bait margin compares a candidate's identity to the nearest labelled IP₃ receptor bait with its identity to the nearest labelled ryanodine receptor bait and assigns it only when one wins by more than 0.10; identity is also recorded over mutually covered columns only. The negative controls, the six ryanodine receptors included, are all rejected.

The architecture rule reads annotation. A record is a ryanodine receptor if it carries PF02026, PF06459, PF21119 or PF00622, and an IP₃ receptor if it carries all of PF08709, PF01365, PF08454, PF02815 and PF00520 and none of those four. Length enters only as support. PF00622 (SPRY) occurs in many unrelated proteins, so its use is conditional on a family seed signature and is audited under that condition. The rule's accuracy is scored against gene symbols it never reads.

The profiles read residues. Seeds were taken from the census with the census's call, one per species per family, disjoint between the two sets, and every selection rule is enforced when the profile is built, so a seed whose call, architecture or length has drifted aborts the build. Seeds were aligned with MAFFT L-INS-i single-threaded, because multithreaded runs of the same seeds gave different alignments, and profiles were built with hmmbuild [16]. A target is assigned when the winning profile scores at least 30 bits over at least 200 match states at $E \leq 1e-5$ and beats the other by more than 10 % of its own score. The 200-state span gate is set between the shortest observed PF08709 (200 residues) and the longest observed SPRY (137); without it a sensitive channel profile calls troponin and helicases ryanodine receptors. The profiles were calibrated against the architecture call on the archived census sequences, with and without the seeds, before any sweep. Both instruments' verdicts are kept on every row and merged by a stated rule: agreement gives a high-confidence call, one instrument speaking gives its call, a disagreement is kept as a conflict, and neither leaves the record unassigned.

Completeness

Enumerated IP₃ receptor records that the vertebrate sweep did not report were each looked up in the 8.8 GB search database by one streaming pass, to separate records never searched from records searched and missed. Iterative searches were run with jackhmmer [17] to convergence or to a ceiling of ten rounds, from three vertebrate-database seeds and from one seed native to each non-vertebrate eukaryotic group, and each round's included set was recorded from the raw log. The kill criterion was evaluated per round on that round's included set, and rounds from its first firing onward were excluded from every merge. Drift was then measured on each finished model as the share of its included targets that neither profile scores, and each rule of the criterion was scored as a classifier of that outcome.

Two sensitivities for every negative claim

Every non-vertebrate search was run once at $E \leq 10$ and filtered at $E \leq 1e-5$, so that the strict and the relaxed sets come from one search. The relaxed panel adds the four family Pfam models, downloaded from InterPro, to the two profiles. That the filtered strict set equals a strict run was tested rather than assumed: no assignment and no gate decision moves, the few domain rows that differ all score at or below 0 bits, and the test fails if it compares fewer than 25 targets. Hits were attributed to proteomes through a measured accession-to-proteome map, because several taxa carry two reference proteomes.

The plant and fungal chase

Every plant and fungal record called IP₃ receptor by either instrument was assigned one of seven verdicts by ordered rules, each writing the number it fired on: unresolved, module-only (under the 200-state floor), contam-

ination suspect (at least 95 % identity over half the query to a protein outside its kingdom, by BLAST [31]), cross-kingdom outlier (80–95 %), no genome backing, fragment (under 2,000 residues), and real gene.

The genome sweep and its controls

Each assembly was searched with miniprot [32] using a panel of 108 baits chosen by seven written rules: every bait's label is its clade band and never a paralogue, length bands and architecture floors are measured per band, and every bait passes a chimera screen against both profiles. The maximum intron length was set per group from sixteen measured gene spans taken from NCBI's own annotations rather than from miniprot, whose setting would otherwise shape the measurement that calibrates it. Where the IP₃ receptor baits won nothing, a family-level tblastn rescue searched the whole assembly. Loci were assigned to a family by the winning bait, graded full, fragment or scrap, and counted as genes after merging split models whose bait spans are complementary and splitting clusters that contain two complete models. The identity floor for a locus was measured against annotation-confirmed loci; because identity did not separate confirmed from contradicted loci outside the vertebrates, every locus was also scored against both profiles.

The positive control for each clade carrying a negative claim was chosen by running six candidate families (MIR, myosin head, E1–E2 ATPase, SMC N-terminal, kinesin and AAA) over that clade's swept proteomes and taking the one present in the most. Each genome receives a verdict: controlled across a kingdom boundary when a control is recovered and baits from another kingdom-level group align across it, controlled by target when a receptor is found, partial when only control fragments are found, or no control bait. An absence is counted only over controlled genomes, in both the numerator and the denominator.

Structures

Candidate structures were enumerated from RCSB on the union of the four family signatures, because RCSB's annotation of human ITPR3 lacks PF08709. References were cryo-EM, full length within the family's band, one per paralogue, with a state panel on the paralogue offering most conformations and negative controls drawn from the benchmark decoy panel and size-matched. Every structure was reduced to one chain. AlphaFold DB coverage was measured as modelled residues over the census length. TM-align [21] was run on all pairs, and a structure was called for a family when its best reference-normalised score cleared 0.5 and beat the other family by a 10 % relative margin. Model confidence was reported per domain, with boundaries transferred by pairwise alignment.

Reproducibility

Every number in this paper is read from a committed table and re-verified against it on every build of the paper. Raw API responses are archived so that each parse re-runs offline, and tool versions are recorded in the committed toolchain manifest.

Figure legends

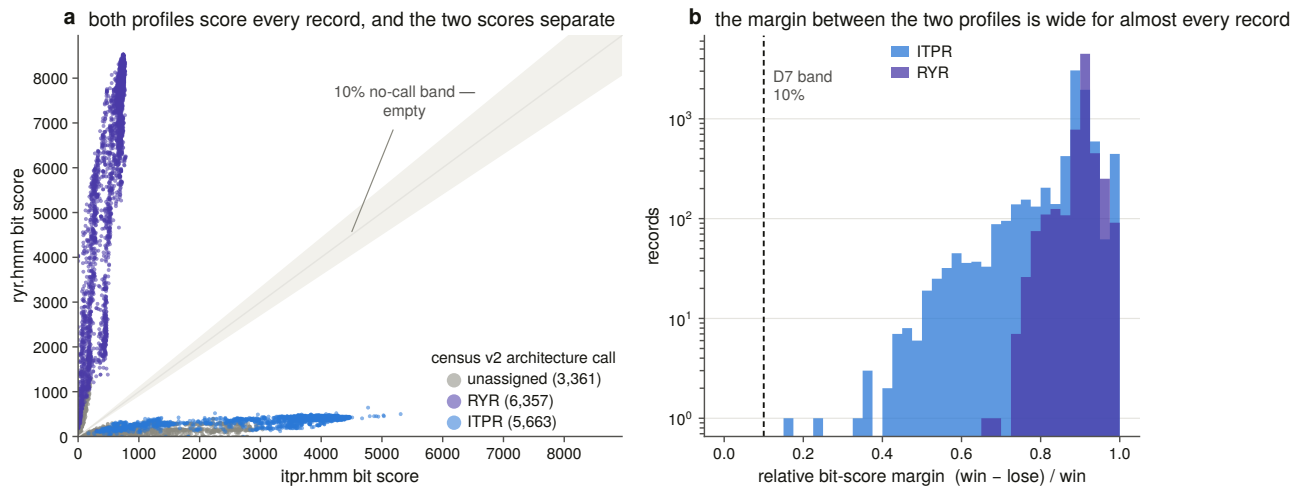


Fig. 1. Two profile models separate the IP₃ receptors from the ryanodine receptors over the whole census, and the band in which neither is called is drawn rather than described. Each record's score against the IP₃ receptor profile is plotted against its score against the ryanodine receptor profile, and the relative margin between them is shown as a distribution with the same band marked. A call requires the winner to clear 30 bits over 200 match states and to beat the loser by more than 10 % of its own score. The margin is relative because bit scores grow with alignable length, so a fixed gap would call every full-length protein and refuse every fragment. Few records fall in the band beside the two dense clouds, which is why the merged call can be treated as settled.

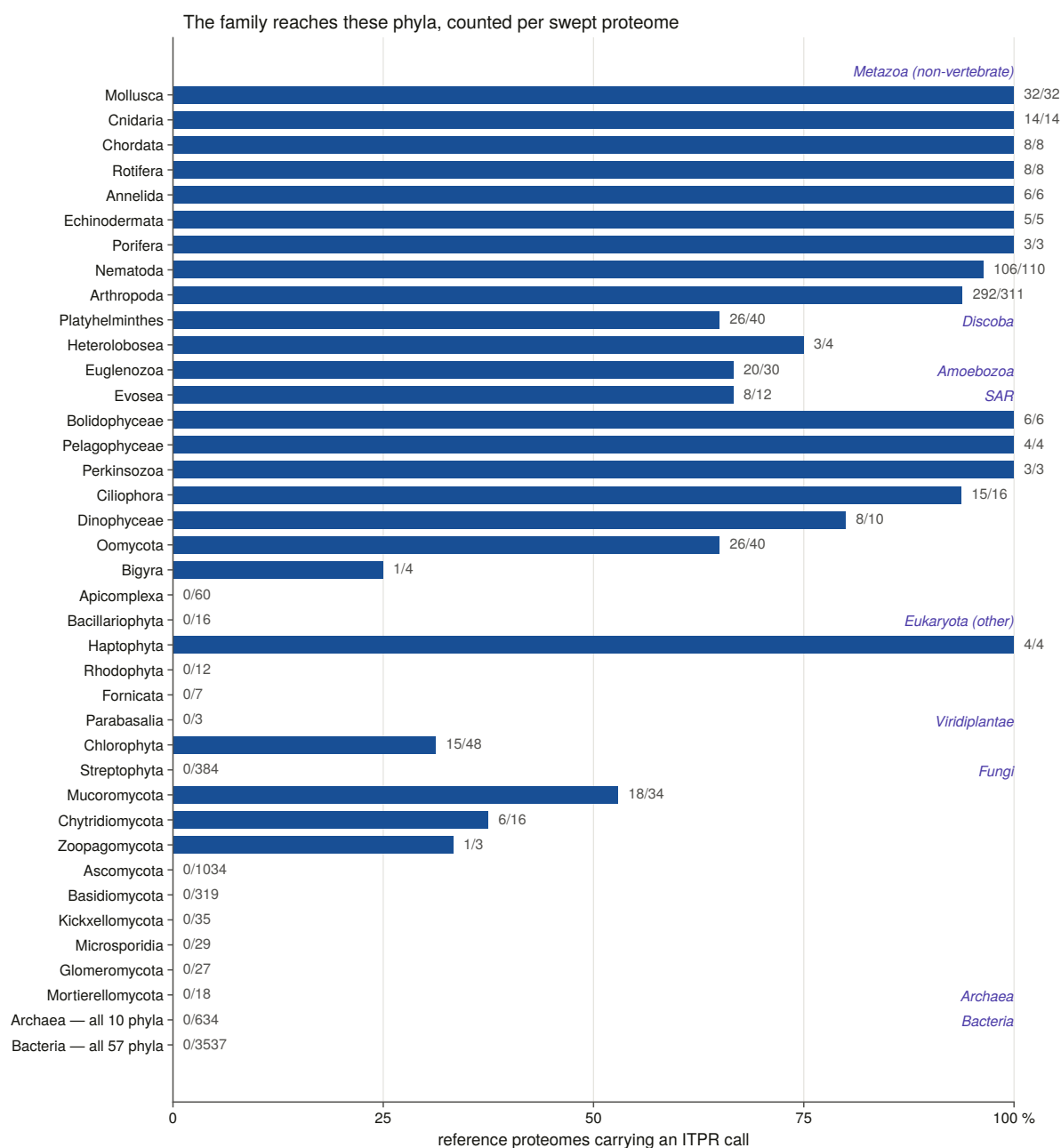


Fig. 2. The family is present across the animals, several protist lineages, the green algae and the early-diverging fungi, and absent from land plants, Dikarya and every prokaryote. Each bar gives the fraction of a lineage's swept reference proteomes that carry at least one IP_3 receptor call, with the count over the denominator printed beside it, and lineages are grouped under the sweep group they belong to. Counting proteomes rather than records keeps a heavily sequenced phylum from outvoting a sparse one. Bars at zero sit in kingdoms where the same search finds the family elsewhere and finds the promiscuous MIR domain in the empty lineages themselves.

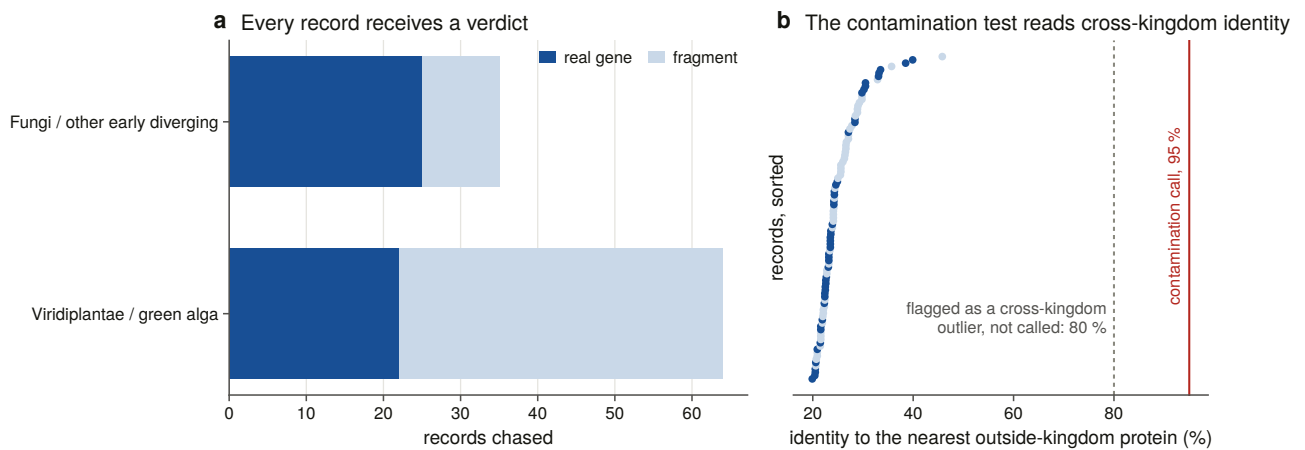


Fig. 3. Every plant and fungal record ends as a real gene or a fragment, and none sits near the identity that marks contamination. Each record's verdict from seven ordered rules is shown with its best identity to any protein outside its own kingdom. An assembly contaminant would be 95–100 % identical to one particular animal, while a deep homologue sits at 20–40 %; every record here falls in the second range, so the green-algal and early-fungal records are genes of those organisms, and the land-plant and Dikarya absences are losses rather than the family's edge.

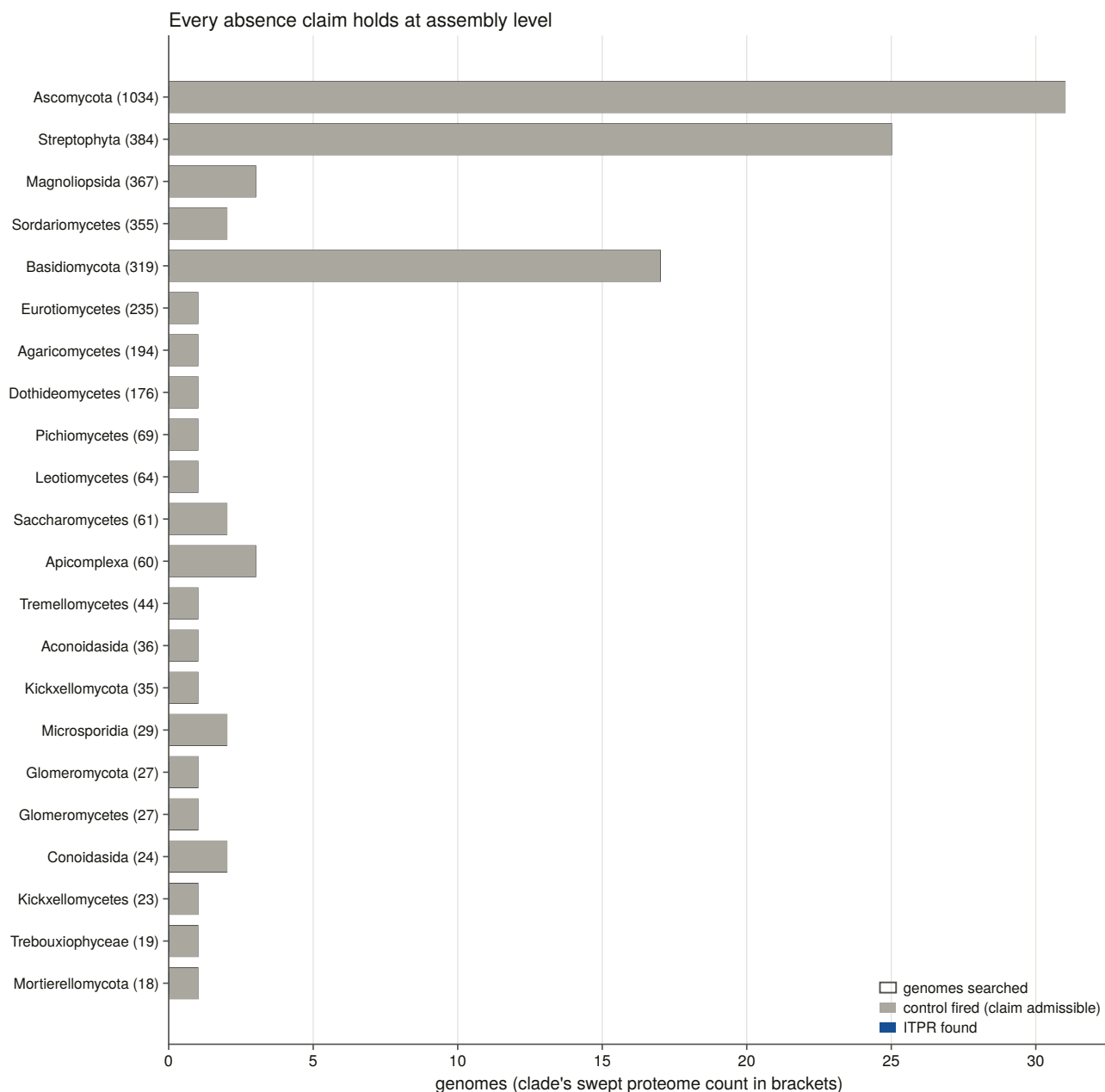


Fig. 4. All 35 clade-level absences hold in genome assemblies that carry a working positive control. For each clade the number of controlled genomes is drawn beside the number searched, and only controlled genomes count on either side. A clade whose control bar were short would be standing on nothing, which is why the two counts are drawn together. No clade has a full-length receptor in any controlled assembly; two land-plant assemblies carry only scattered tblastn traces.

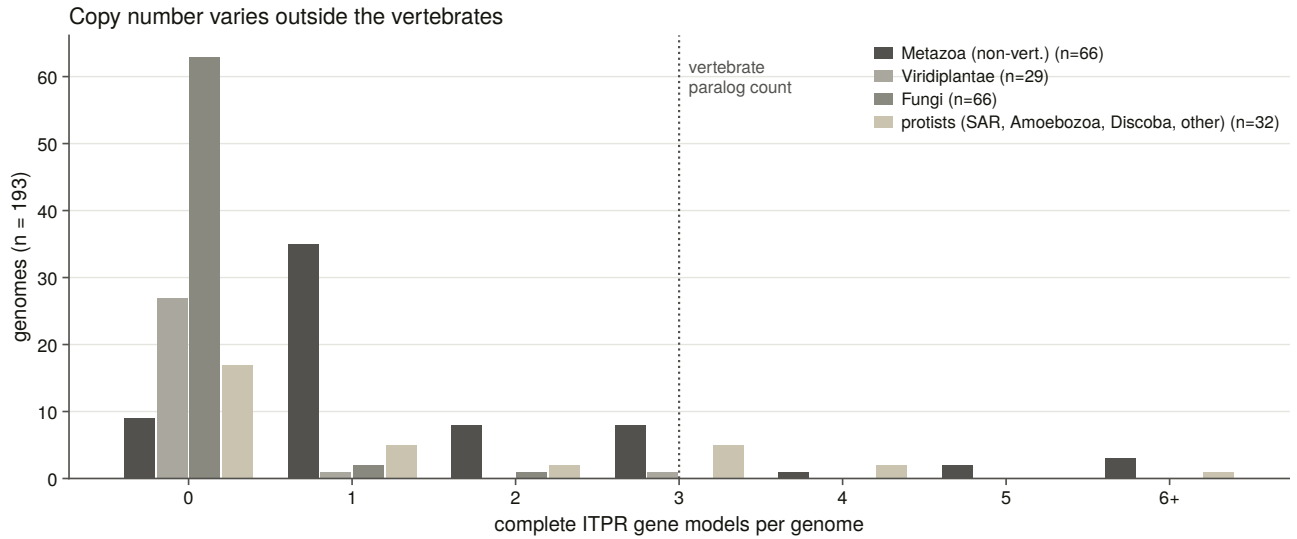


Fig. 5. Copy number outside the vertebrates runs from zero to eighteen, and the vertebrate paralogue count of three is drawn as a reference line rather than a category. Each point is one genome's count of complete receptor gene models, grouped by taxonomic group. A flatworm carries eighteen and a ciliate thirteen, so expansion in this family is not specific to the vertebrates.

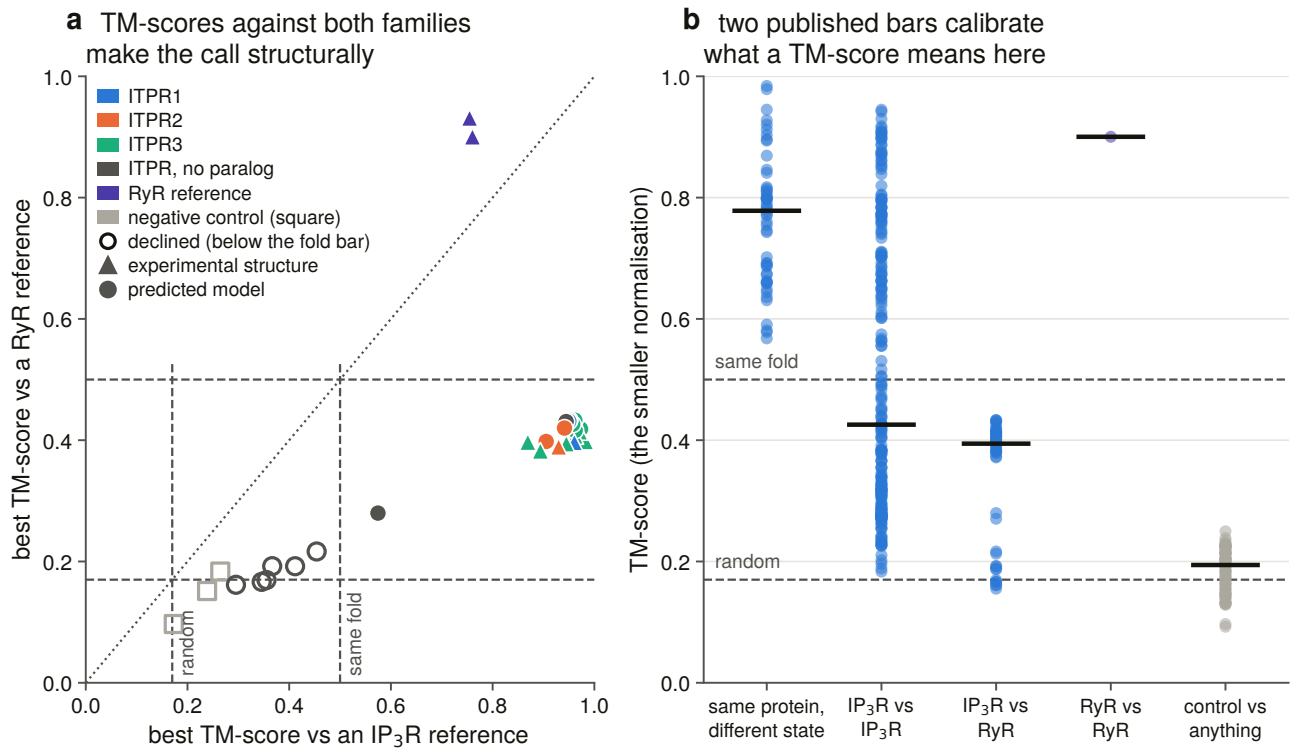
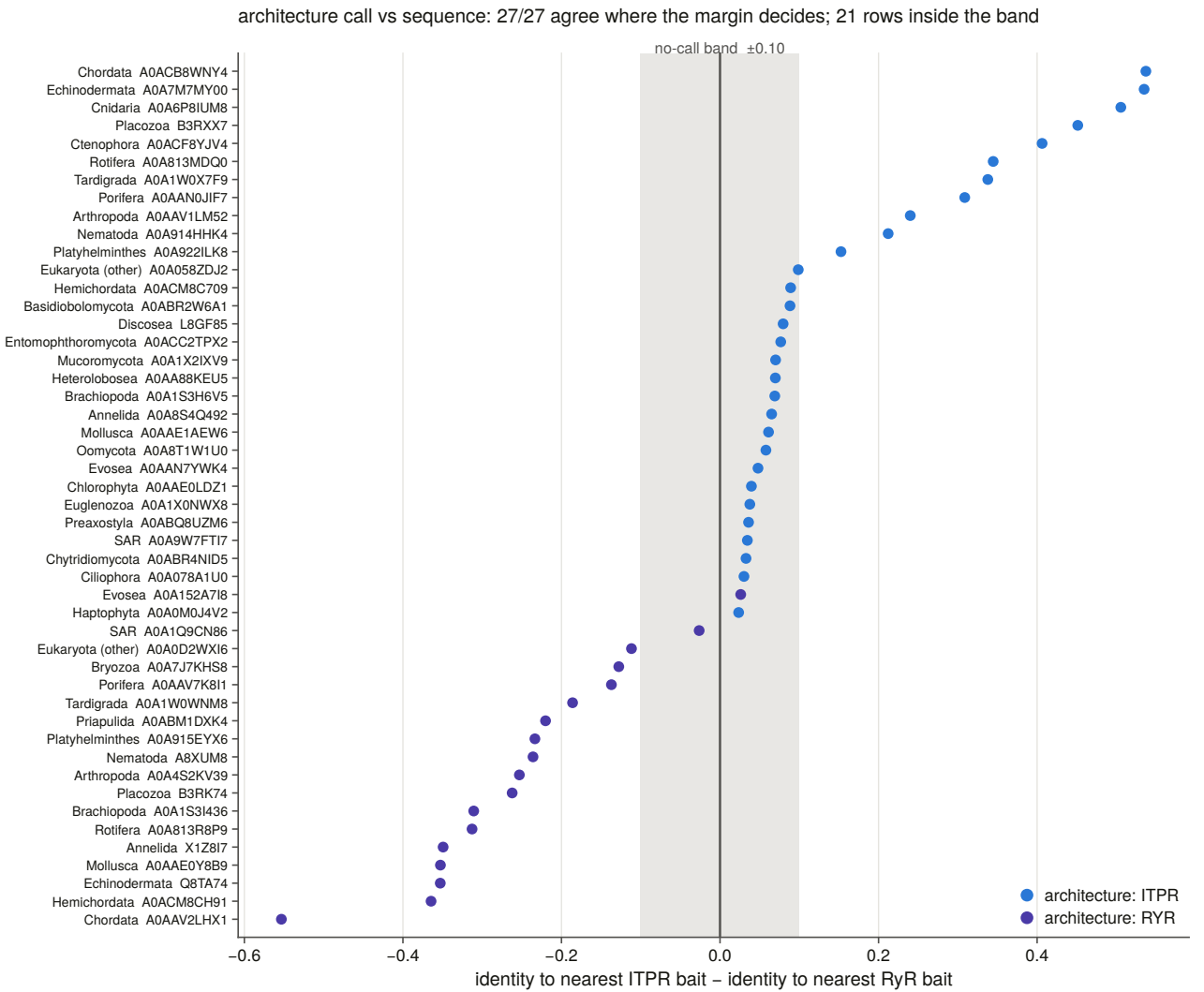


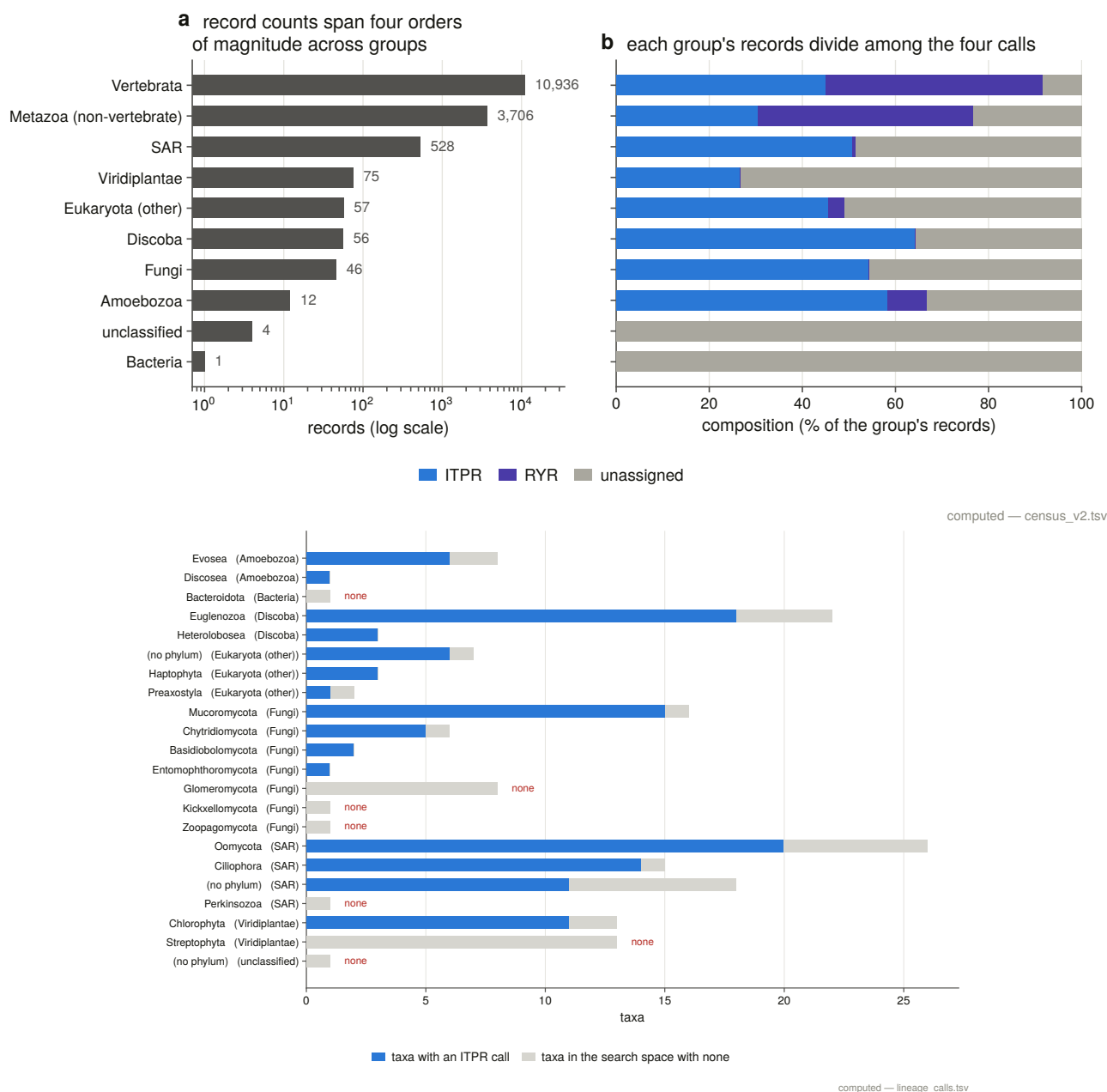
Fig. 6. The fold separates the two families as the sequence census does, on a scale calibrated on this panel before any structure was called. **(a)** Each structure's best TM-score against the IP₃ receptor references is plotted against its best against the ryanodine receptor references, both normalised by the reference. **(b)** Every class of structure pair is drawn against TM-align's published random (0.17) and same-fold (0.5) bars: negative controls stay below the same-fold bar, and different conformations of one paralogue score higher than different paralogues. Recovering the family split from coordinates alone matters because the two families share every diagnostic domain.

Extended Data

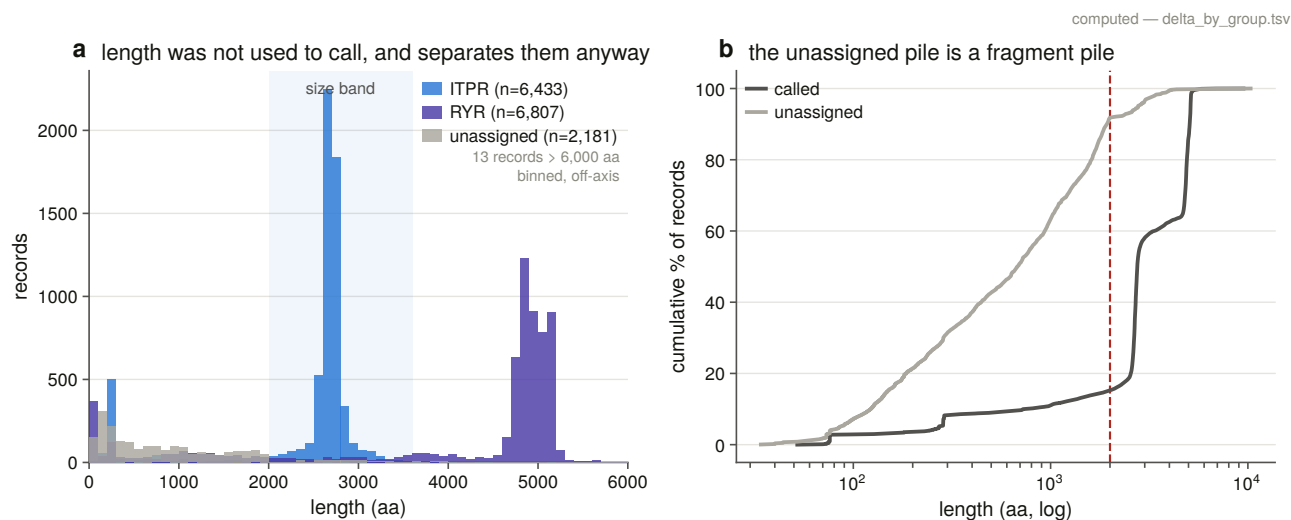
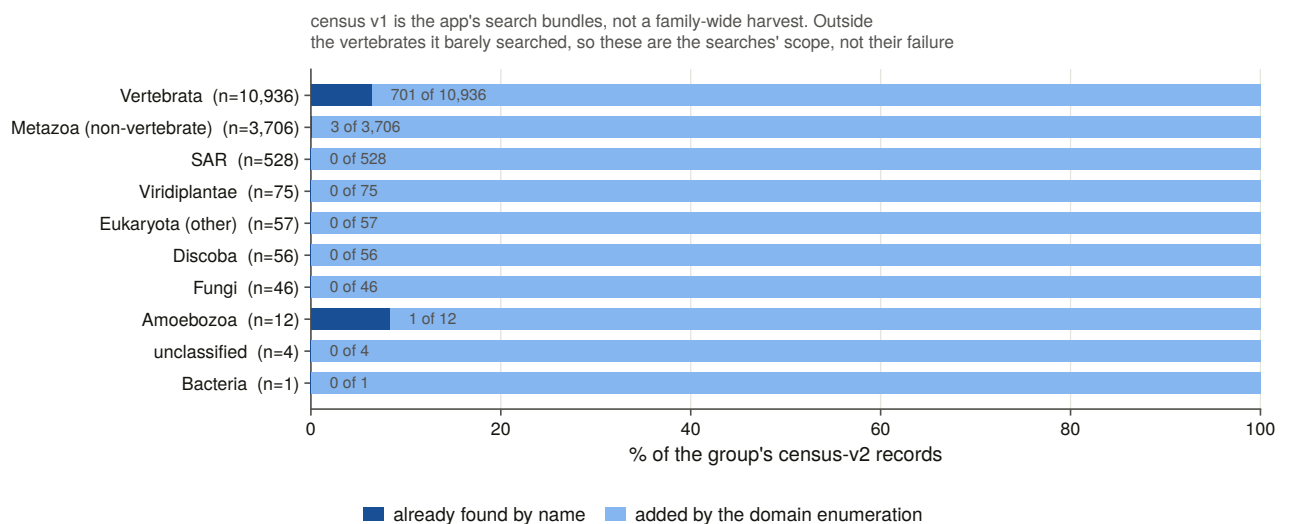


computed — sequence_check.tsv

Extended Data Fig. 1. The labelled-bait margin leaves an empty gap between the two families across the whole census. Each record's margin between its identity to the nearest labelled IP₃ receptor bait and to the nearest labelled ryanodine receptor bait is drawn with the no-call band marked. No record falls between the narrowest correct call and the narrowest correct rejection, so the 0.10 margin is not a compromise between sensitivity and specificity.

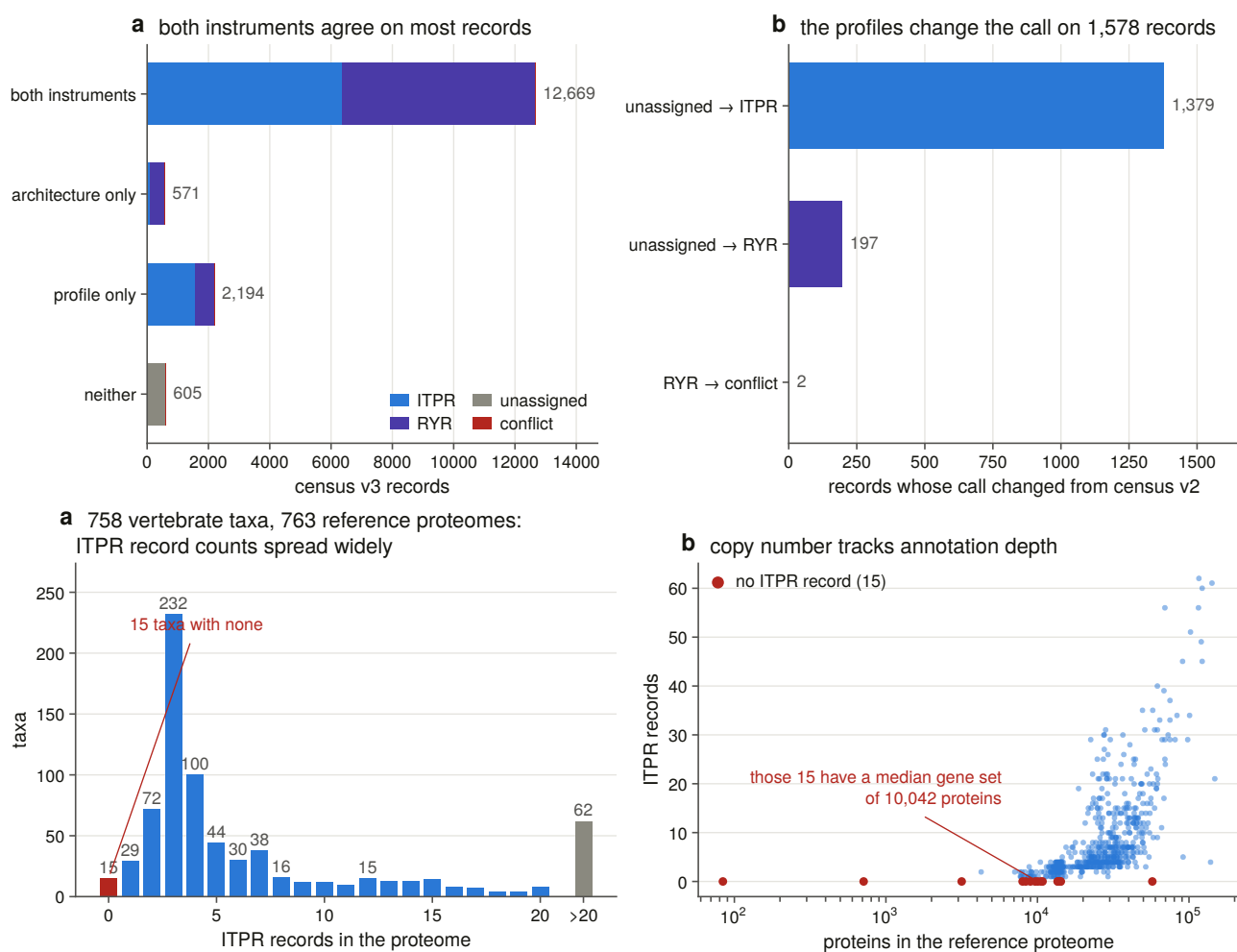


Extended Data Fig. 2. The search space was enumerated by three signatures because no single one reaches the whole family. **(a)** The size of each signature's protein set and of every overlap between them is drawn, with each region broken down by what the architecture rule calls it; a census built on PF08709 alone would miss 2,914 proteins. **(b)** The census is broken down by lineage with both family calls; the vertebrate bar is the tallest because sequencing has concentrated there, which is why range is measured in proteomes rather than records.

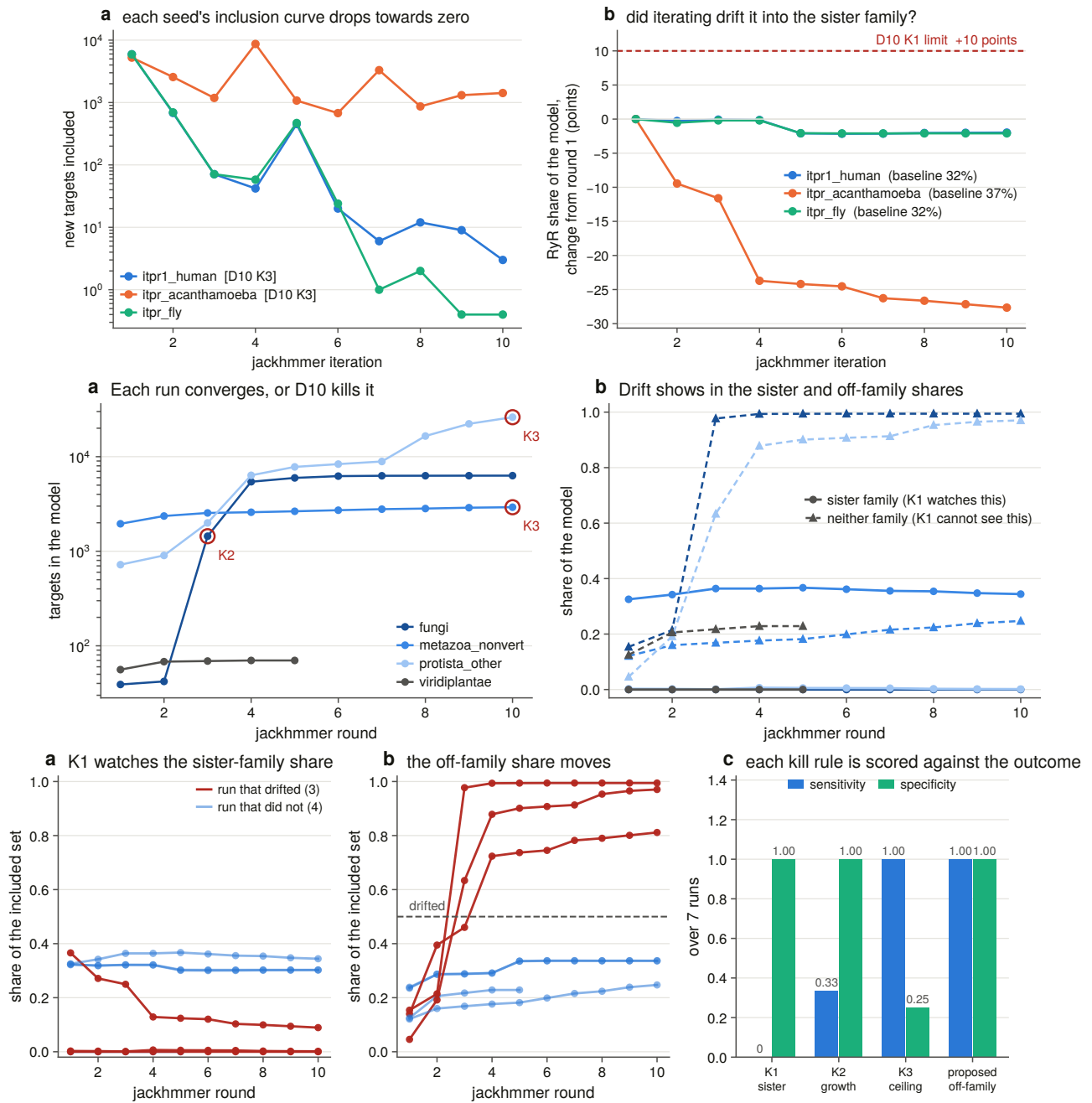


computed — census_v2.tsv

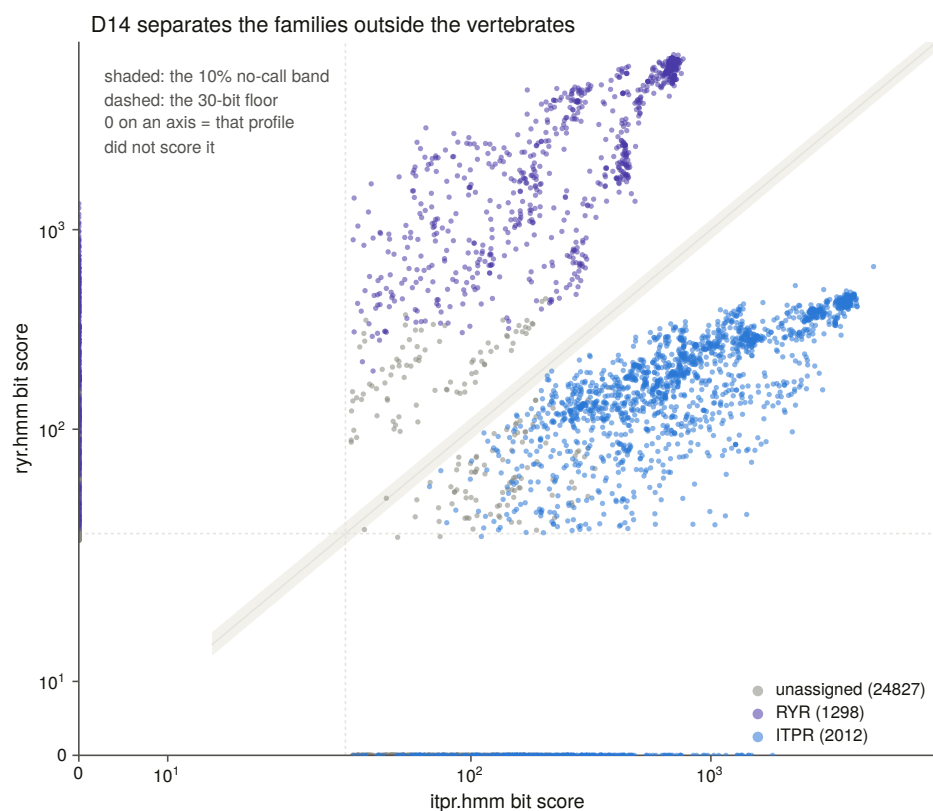
Extended Data Fig. 3. An exhaustive domain enumeration found what a search by name had not, and length alone could not have made the call. **(a)** Each taxonomic group's records are split into those a name-based search had already returned and those the enumeration added, and outside the vertebrates the name-based share is close to zero. **(b)** The two called families separate by length, while the records neither rule can call are short fragments far below both.



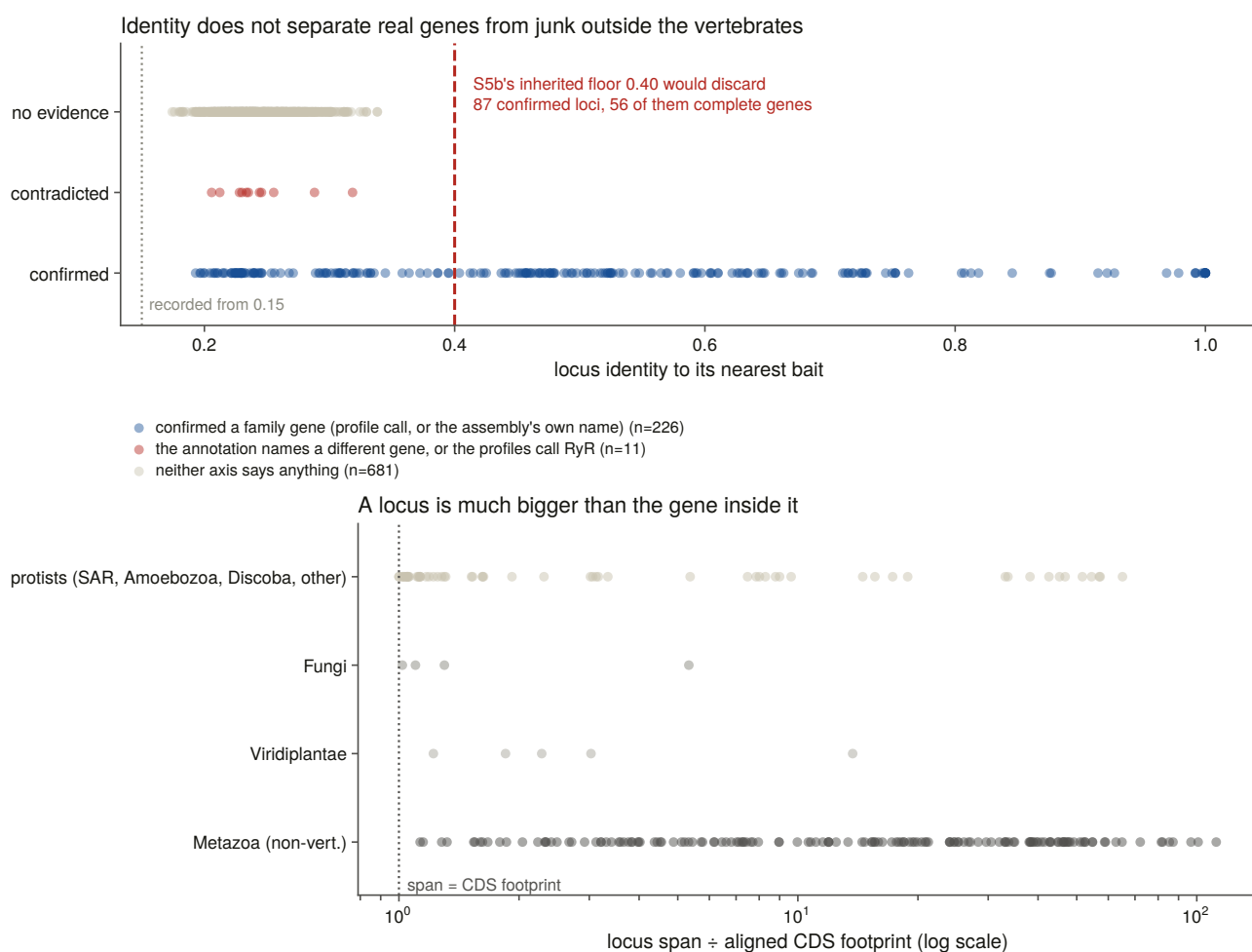
Extended Data Fig. 4. Two instruments reading different evidence speak for most of the census, and gene-set size confounds per-proteome copy number. **(a)** Records are split by whether the architecture rule, the profiles, both or neither call them, and the profiles' contribution is almost entirely records the architecture rule could not decide. **(b)** Receptor calls per vertebrate proteome are plotted against the size of its gene set, and the fifteen proteomes with no call sit among the smallest gene sets.



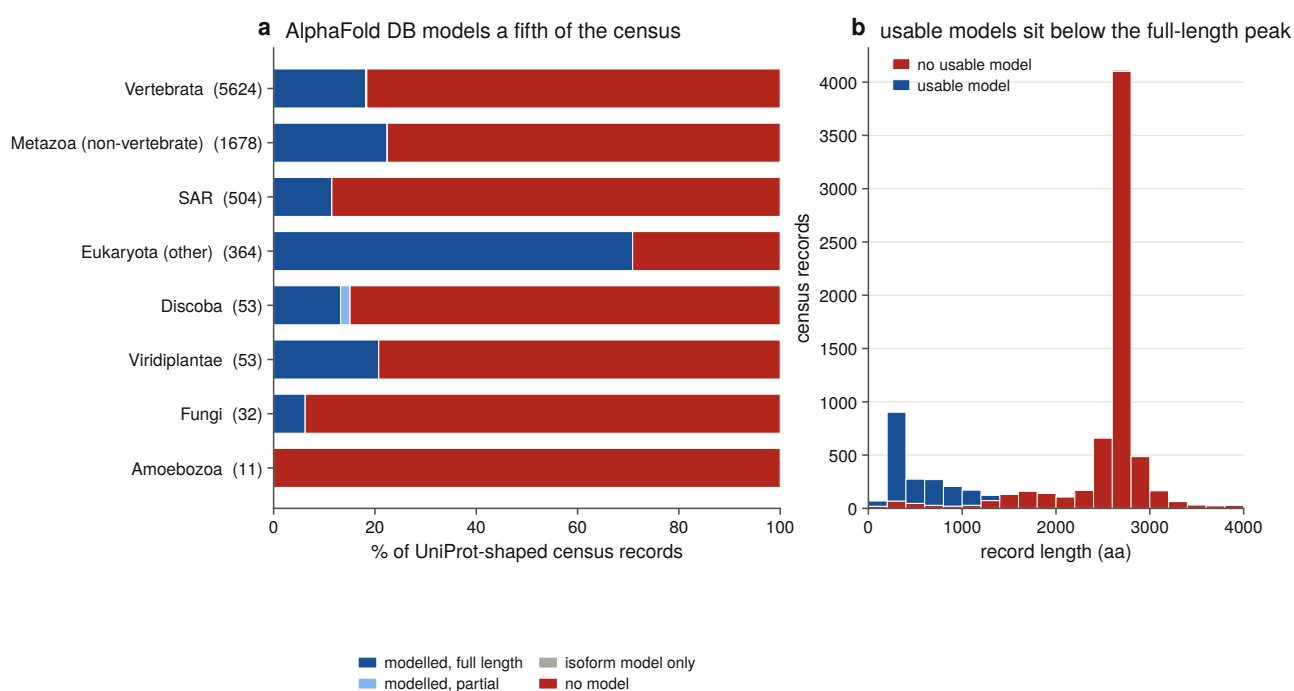
Extended Data Fig. 5. Iterative searches from distant seeds recover the same family and differ only in what is not family. (a) The three vertebrate-database runs, seeded from a human, a fly and an amoebozoan receptor, are traced by new targets per round and by their ryanodine receptor share. (b) The same traces are shown for the runs seeded inside each non-vertebrate eukaryotic group. (c) Each kill rule is scored as a classifier of drift measured on the finished model: the rule written for this family's hazard fires on none of the drifted runs, and the same threshold applied to the off-family share separates all seven.



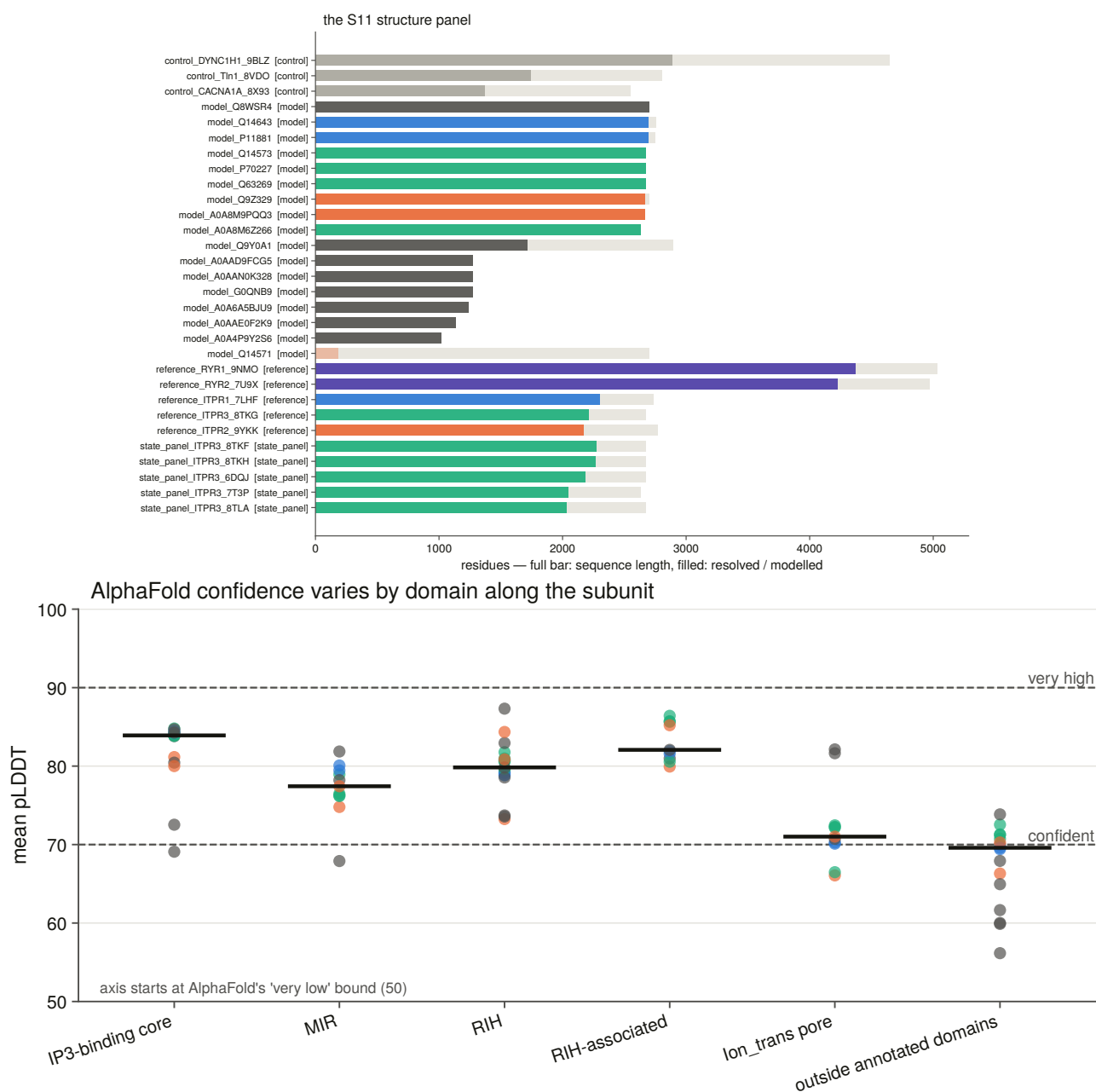
Extended Data Fig. 6. The two-profile separation holds outside the vertebrates. Each target scored by either profile in the non-vertebrate sweep is plotted by its two scores, with the no-call band drawn; of 2,769 targets scored by both profiles above the floor, one falls in the band.



Extended Data Fig. 7. Two genomic thresholds were measured for this scope because neither transfers from the vertebrates. **(a)** The identity of every recorded locus cluster, with loci confirmed and contradicted by the assembly's own annotation shown separately; identity does not separate them, so a second axis scores every locus against both profiles. **(b)** Locus span against the coding footprint inside it, by group; the ratio differs between metazoan and protist genomes, so the intron setting and the contiguity bar are set per group.



Extended Data Fig. 8. AlphaFold DB holds almost no full-length model of this family. **(a)** The modelled fraction of the census is drawn per taxonomic group. **(b)** Modelled and unmodelled records are drawn against record length; the models sit mostly below about 1,300 residues, while the peak at about 2,700, a full-length subunit, is almost entirely unmodelled.



Extended Data Fig. 9. Structural evidence exists for a small part of the family. (a) Every structure in the panel is drawn with the length its record claims and the length it resolves. (b) Prediction confidence is shown per domain in subunit order, with the IP₃-binding core the best-modelled domain and the pore the worst of the named ones.

Data availability

Every table and figure this paper reports is committed in the project repository under `results/` (the benchmark, the census editions v2, v3, v5 and v6, the proteome sweeps, the genome sweep and its bait panel, the structural panel and the iterative-search audit), and `papers/range/deposit_manifest.tsv` lists each deposited file with its SHA-256. Bulk inputs (reference proteomes, genome assemblies, structure files and archived API responses) are not redistributed; each is addressed by a committed accession manifest, and `papers/range/deposit_notes.md` gives the command that regenerates it from its public source.

Code availability

The analysis code is in the repository's `scripts/` directory, one script prefix per analysis stage, and `python scripts/s26_assemble.py --paper range` rebuilds this paper from the committed tables, re-verifying every load-bearing number against the table it comes from.

Author contributions

George Dickinson conceived and directed the project and reviewed the work. The analysis code and the text were written by Claude (Anthropic) under his direction; Chapter 16 of the accompanying thesis describes how the work was carried out.

Competing interests

The author declares no competing interests.

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Citations are rendered from `results/s0_baseline/references.tsv`, numbered in order of first appearance.

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